



Critical Design Review (CDR)

Self landed CANSAT and Canards controlled active stability system

Konlawat Phaikaew



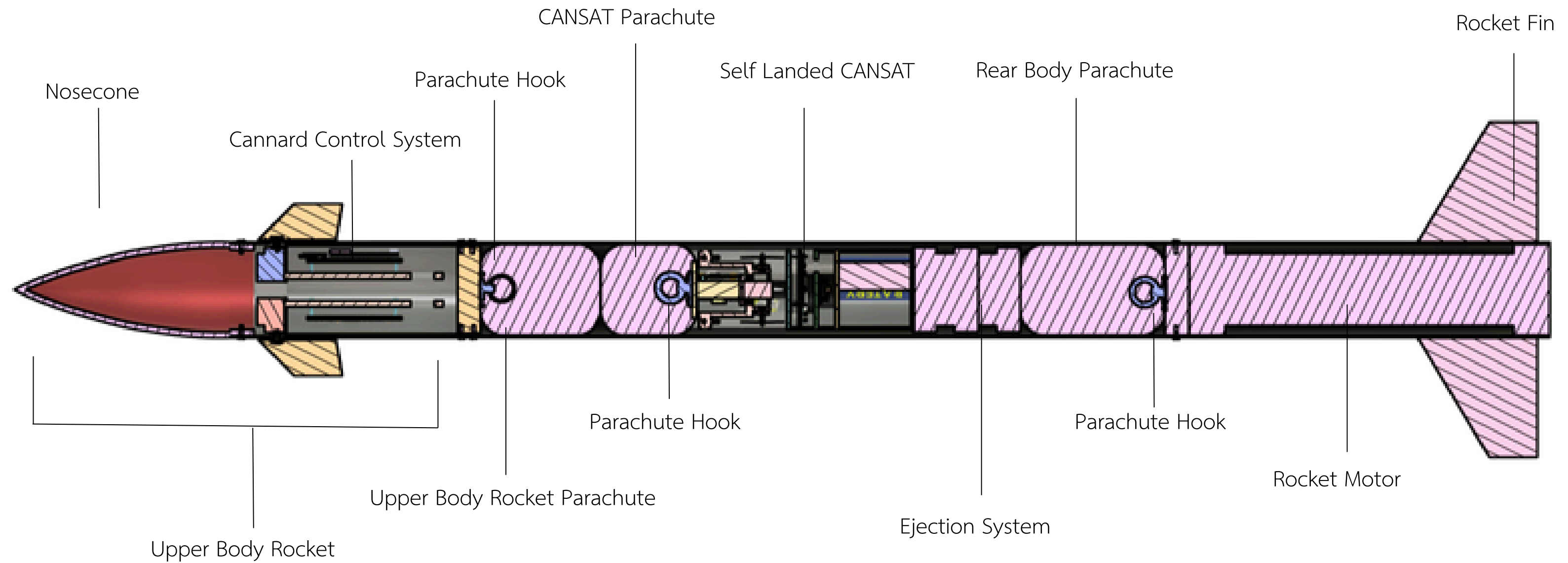
Main Objectives

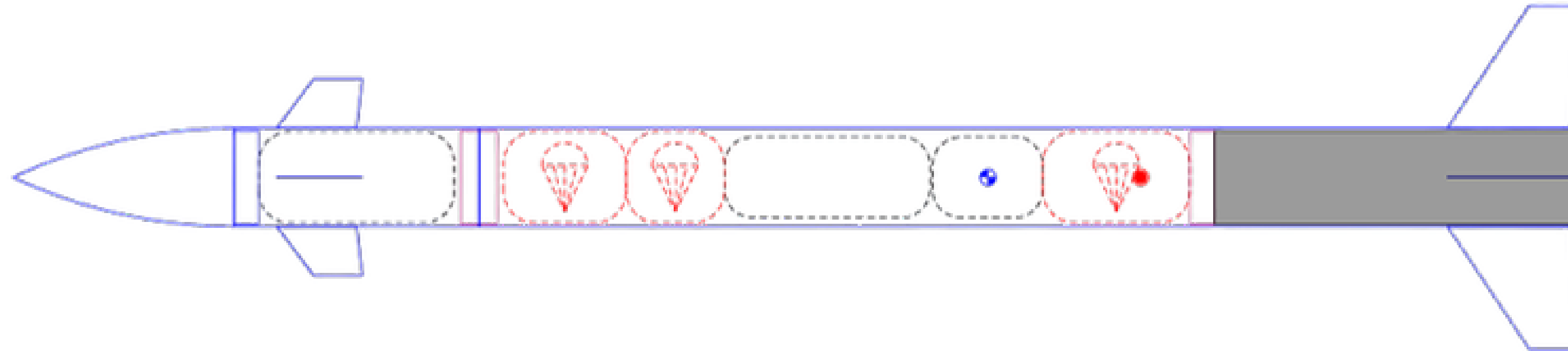
1. The CanSat can transmit data back to the ground station using the LoRa communication system on the 923.525 MHz frequency band and the RadioLib library.
2. Can confirm the CanSat's position with the ground station during the mission.
3. Can measure air pressure, temperature, and humidity.
4. Can operate in the air for no less than 90 seconds.
5. Can measure acceleration during launch and landing.
6. Can carry at least 1 quail egg with the egg remaining intact.

External Objectives

1. Use canards to steer the CanSat toward a specific GPS landing coordinate by adjusting pitch and roll during descent using MATLAB.
2. Activate drone motors at low altitude to neutralize drift and ensure a soft, pinpoint touchdown.







Rocket

Stages: 3

Mass (with motors): 2223 g

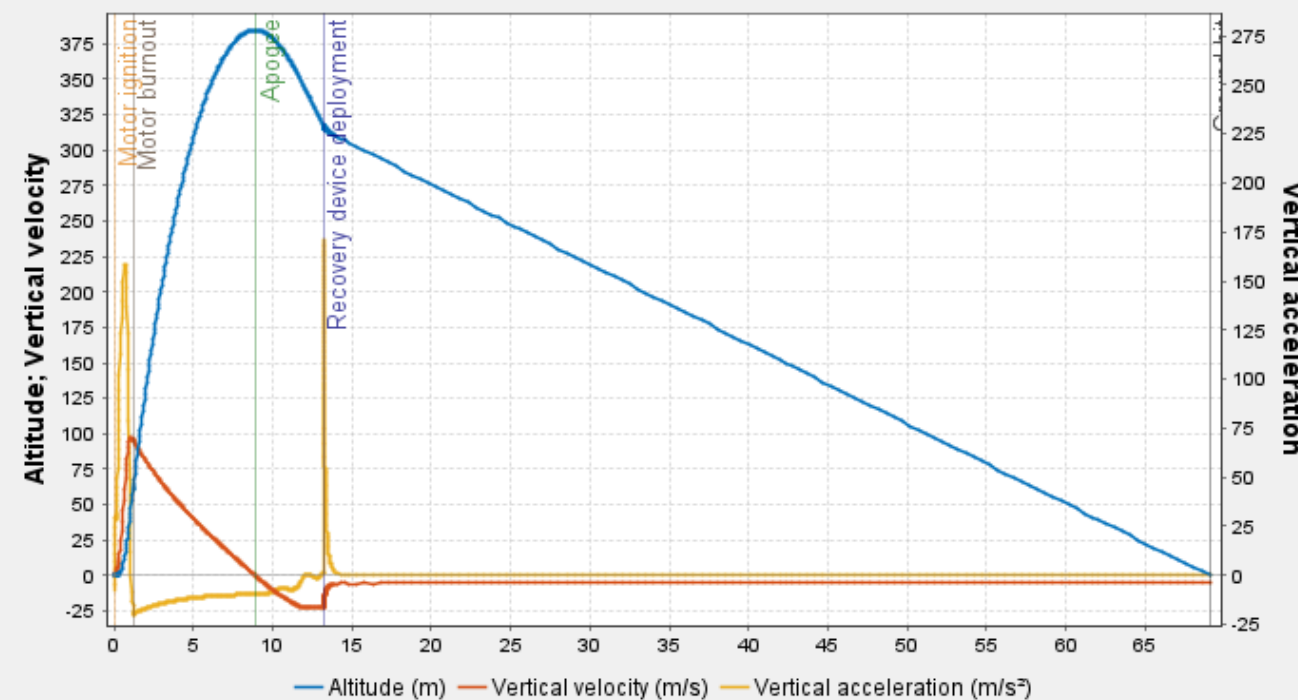
Stability: 1.55 cal / 9.72 %

CG: 79.5 cm

CP: 92 cm

Simulation 3

Vertical motion vs. time



; ; H190-12

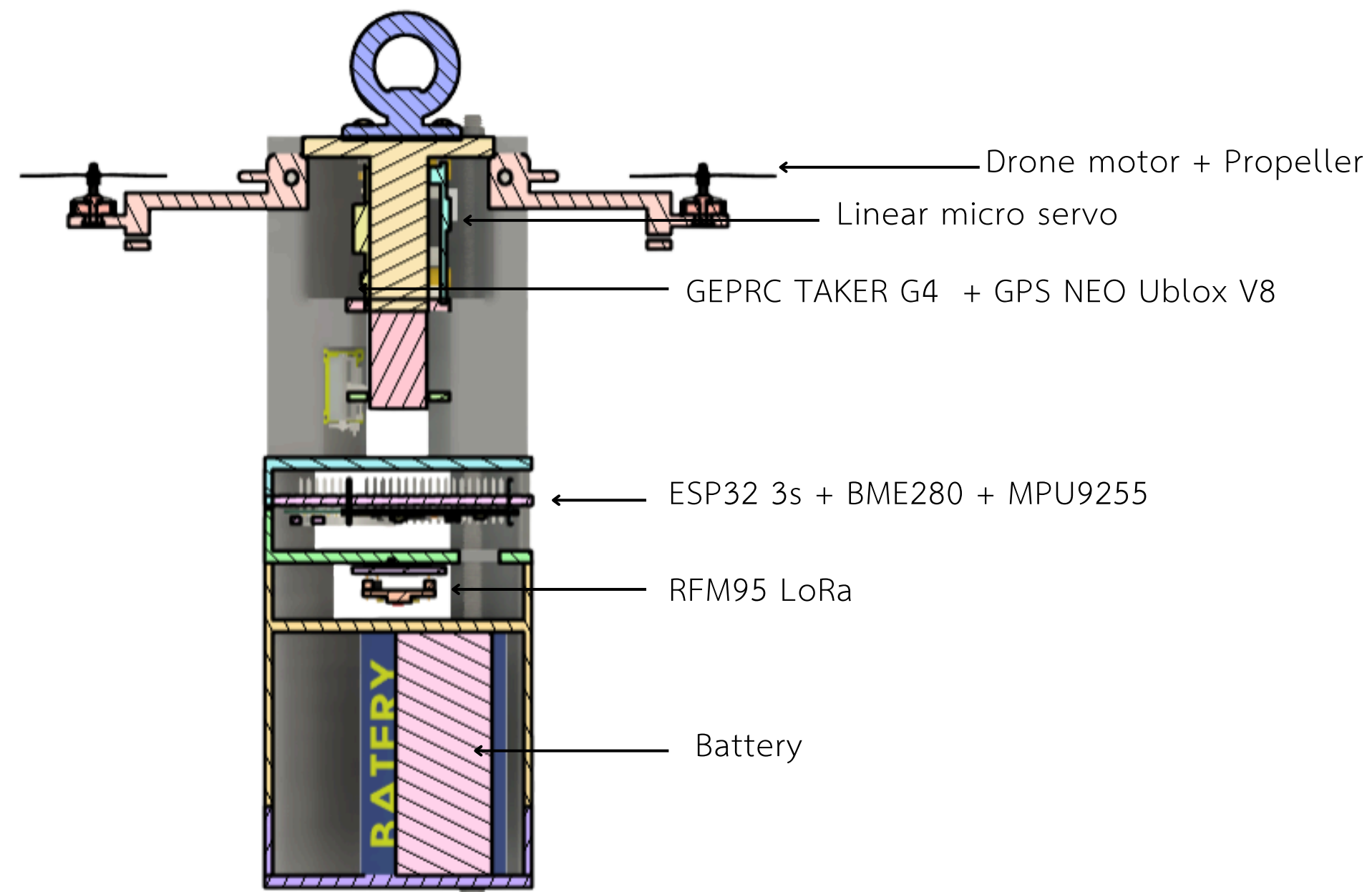
Altitude	385 m
Flight Time	69 s
Time to Apogee	8.88 s
Optimum Delay	N/A
Velocity off Pad	11 m/s
Max Velocity	97.3 m/s
Velocity at Deployment	27.1 m/s
Landing Velocity	5.59 m/s

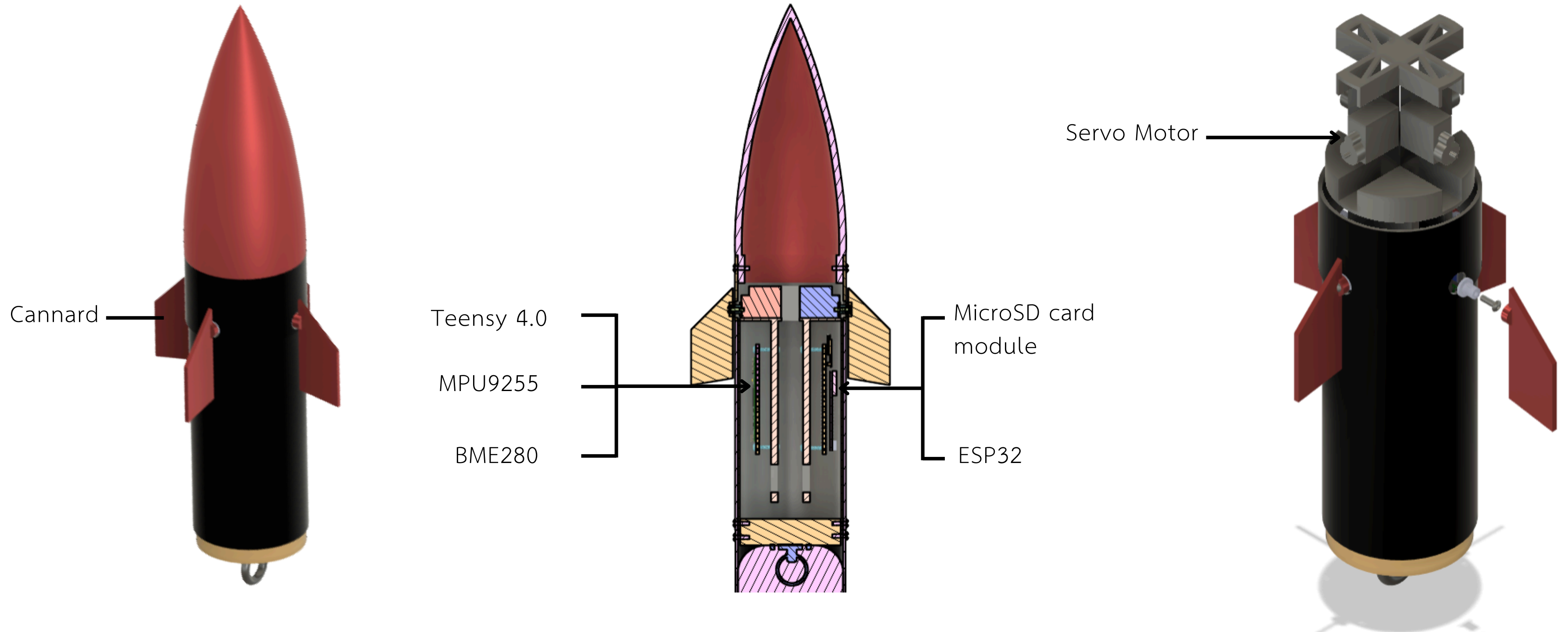
Motor	Avg Thrust	Burn Time	Max Thrust	Total Impulse	Thrust to Wt	Motor Wt	Size
H190	219 N	1.08 s	363 N	238 Ns	10.02:1	210 g	76/300 mm

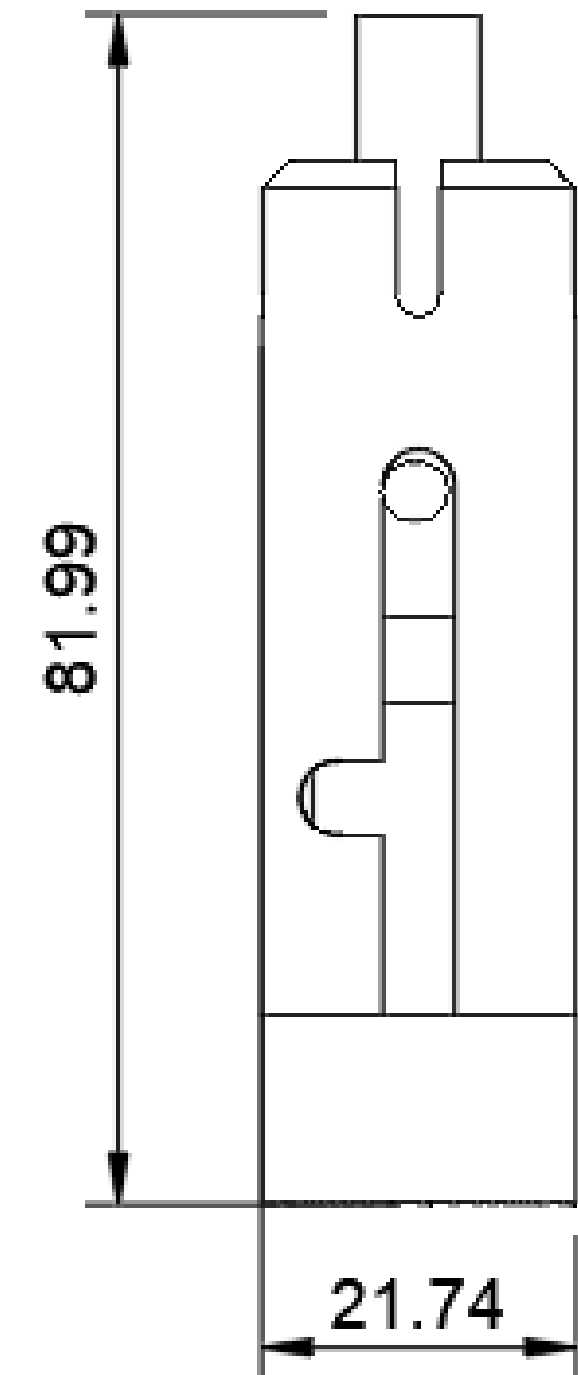
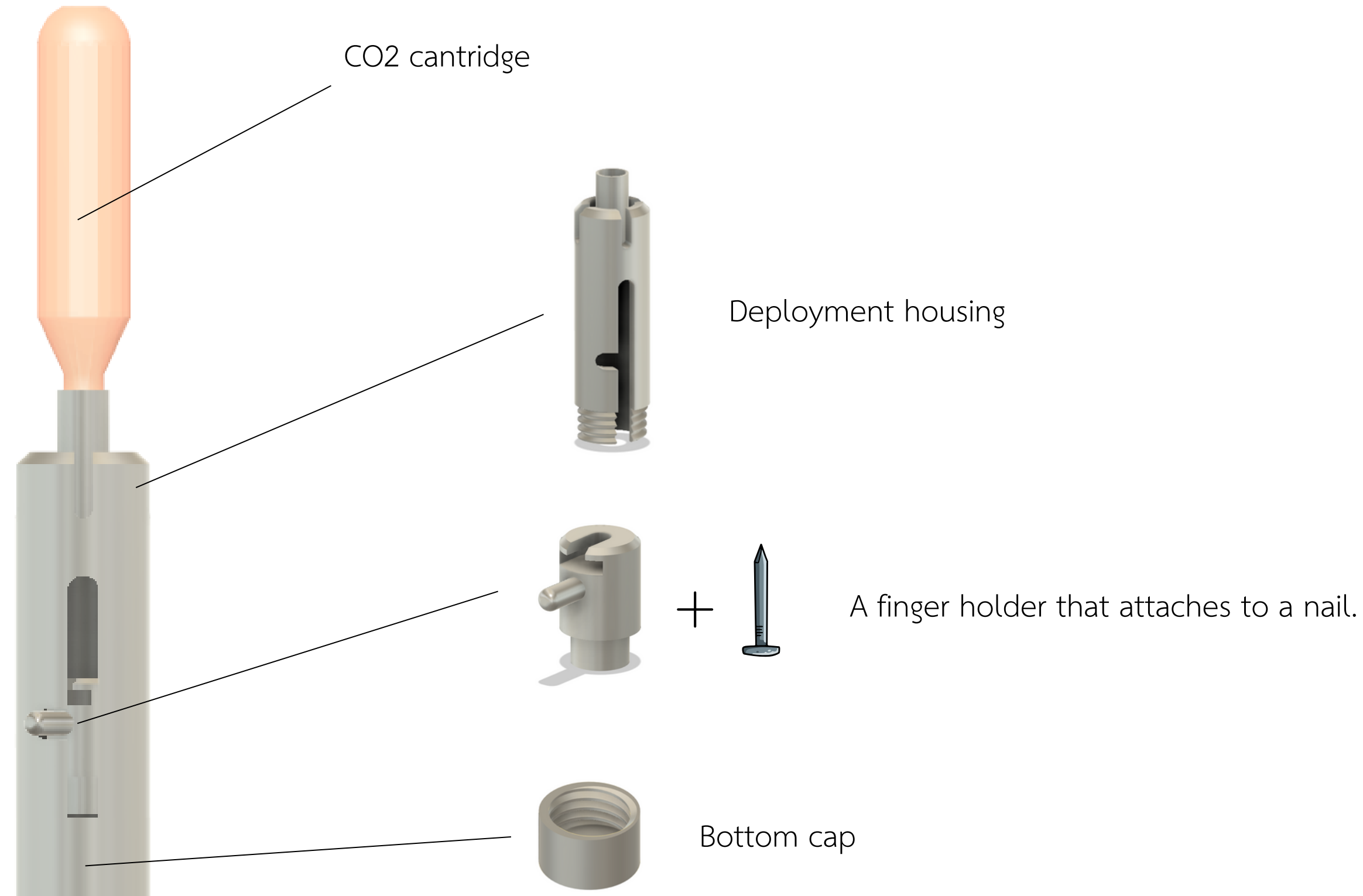




Payload Configuration

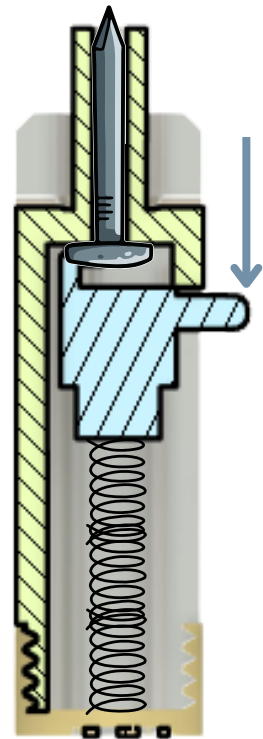




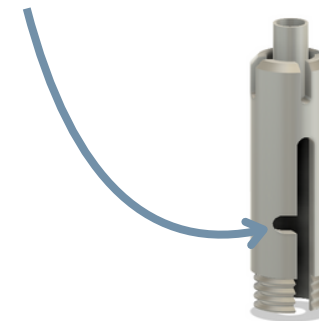
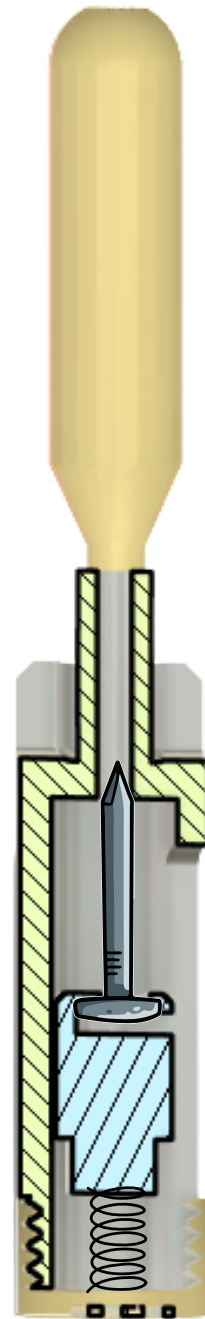




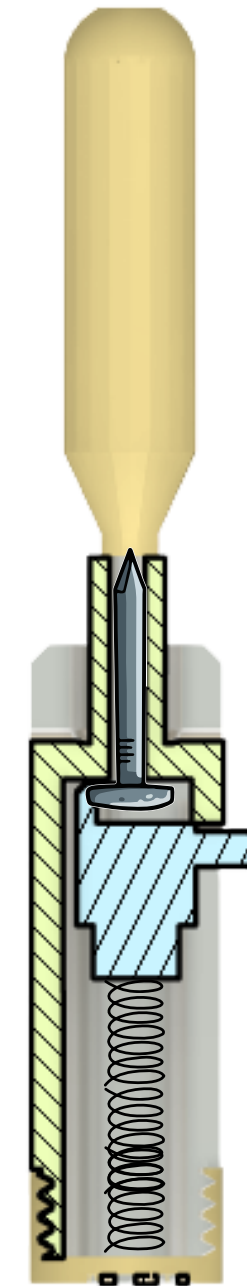
1. Push a finger holder down as far as possible.

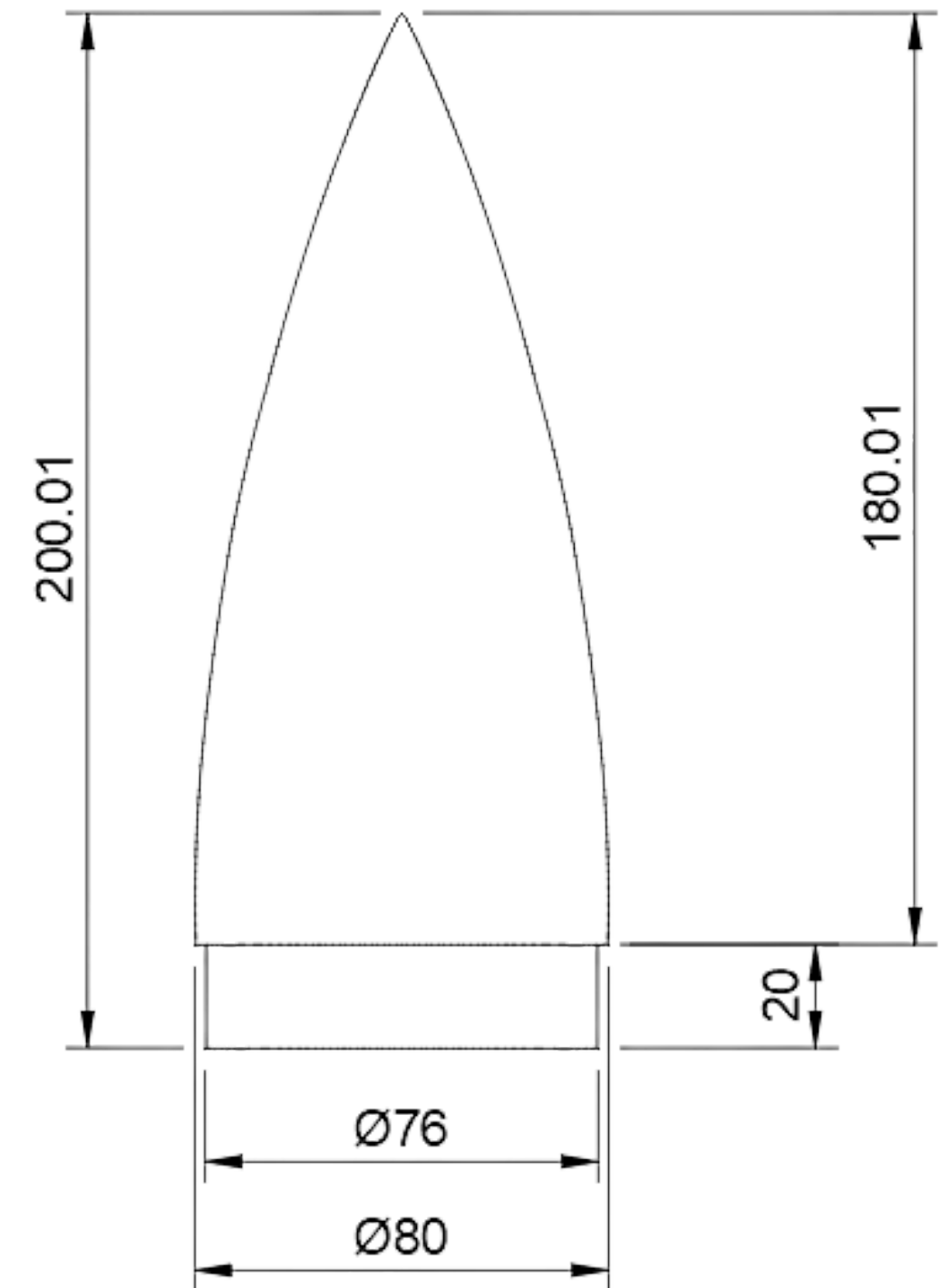
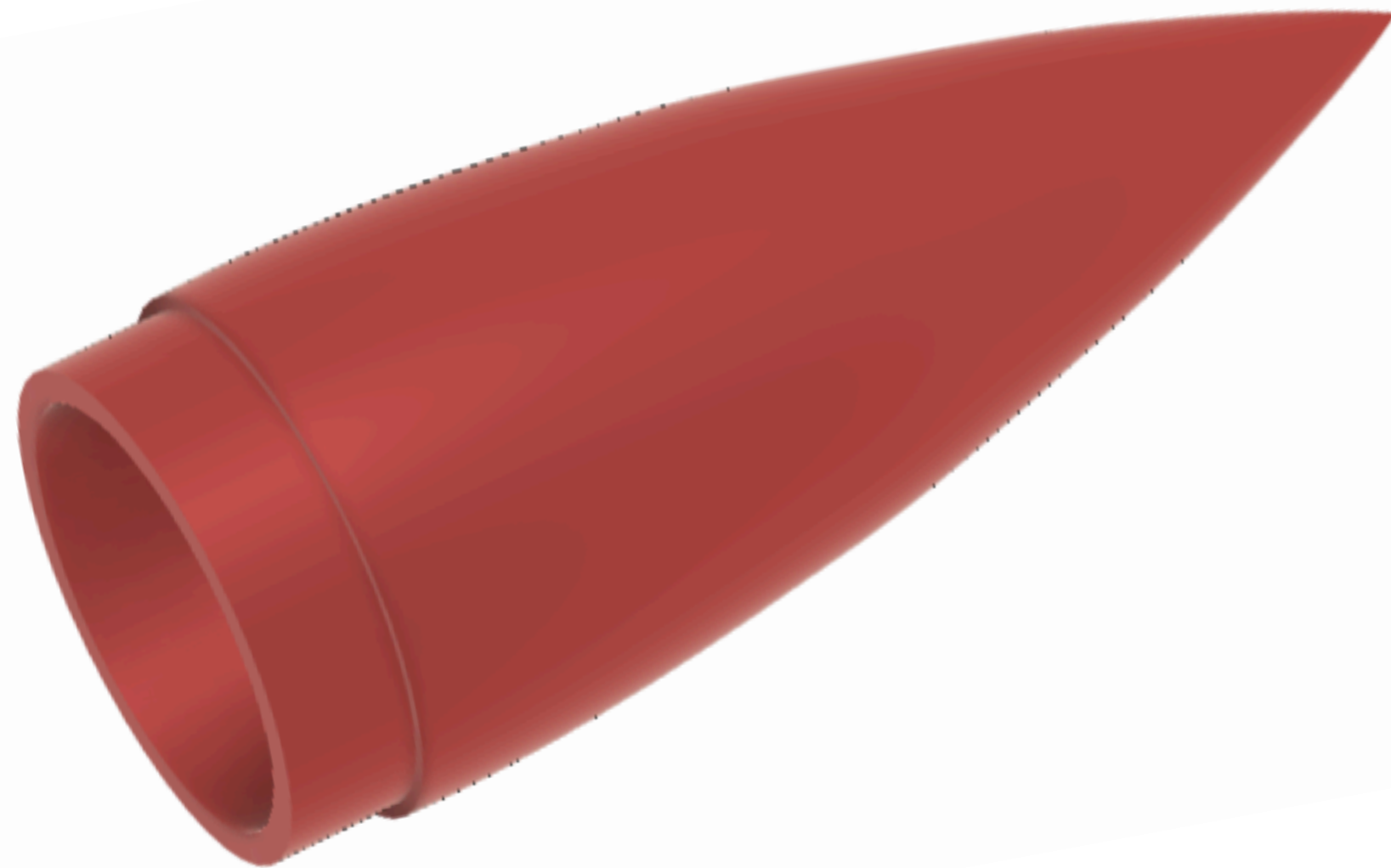


2. Lock the finger holder into the locking slot to store spring potential energy.



3. The servo will release the finger holder at the apogee, causing the spring to snap the finger holder (which is attached to a nail) to puncture the CO2 cartridge.







Sensor Subsystem Overview

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Sensor	Model	Function
Temperature	BME280	Measures temperature inside CANSAT
Pressure	BME280 / MS4525DO	Measure pressure values throughout descent
Accel / Gyro	MPU6050	Measures tilt and rotation angles in relation to x-plane
GPS	GPS Ublox NEO-6M	Measures longitudinal and latitudinal position of CANSAT



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Sensor	Model	Function
Tranciever	RFM95 LoRa	Transmits live sensor data
Flight Controller	EPRC TAKER G4	Stabilize the craft and control the landing motors
Cannard Controller	Teensy 4.0	Data processor and UART sender for complex MATLAB processing
Data Sender	ESP32 S3	Log data to the SD card and handle wireless telemetry